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Simplifying Cloud Security to Ensure a Consistent, Robust & Reliable Security Stance

Cloud Firewall Architecture-HeroMoto

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Feature of Fortinet Firewall on AWS Cloud

- Protects against malware, exploits, and malicious websites in both encrypted and non-encrypted traffic.
- Centralized Access control for multiple AWS Accounts & VPCs.
- Provide security from malicious payload using build-in IPS(Intrusion prevention system) Functionality.
- Geo-location based traffic filtering
- Category based URL filtering for all AWS Accounts.
- Granular policy enforcement based on Application inside traffic.
- Prevent and detect against known and unknown attacks using continuous threat intelligence
- Multi Availability Zone deployment using AWS Gateway Load Balancer.
- Site-to-Site VPN traffic Inspection.
- Remote access VPN.

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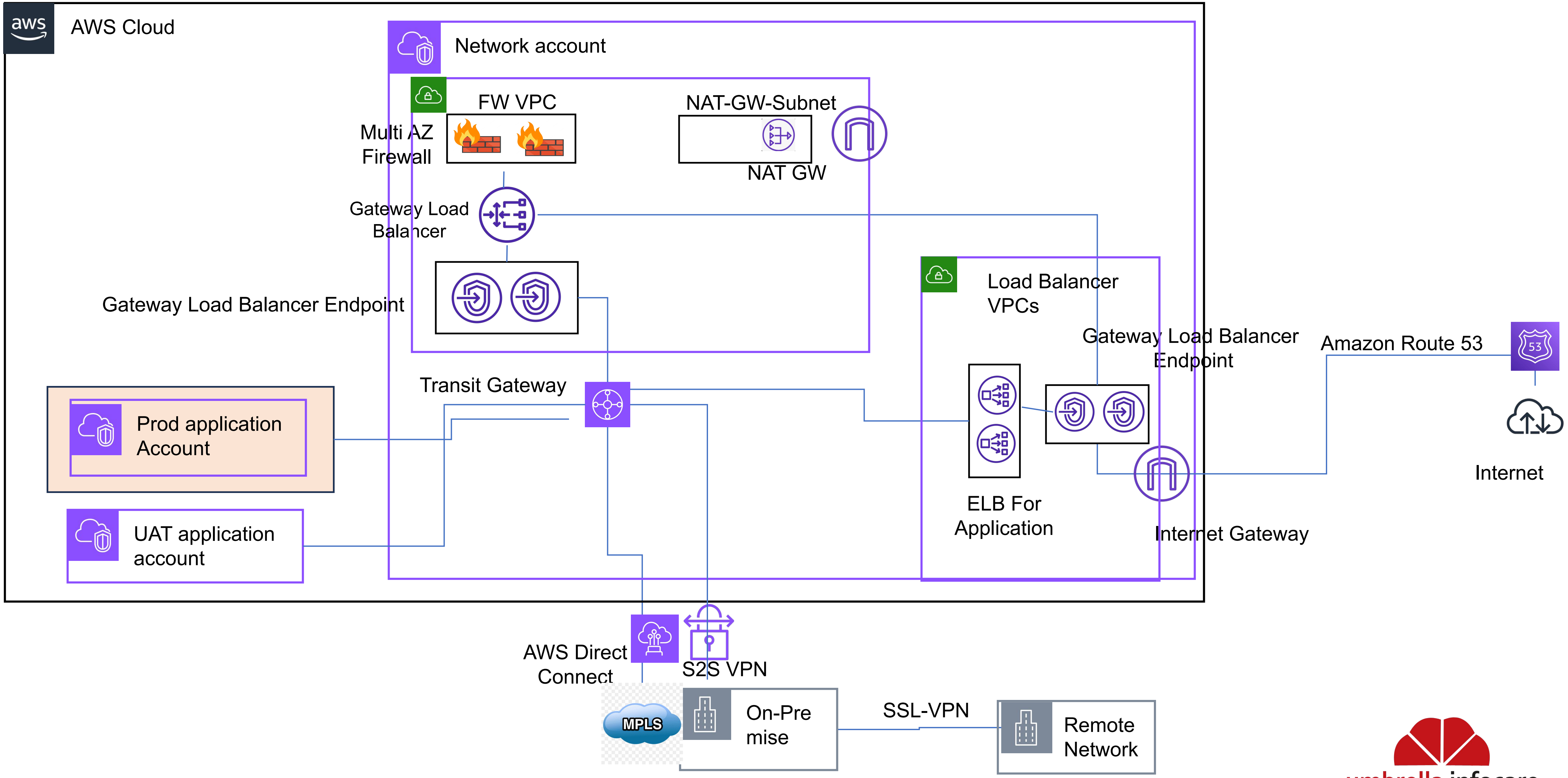
Architecture (ELB\GWLBe in dedicated VPC)



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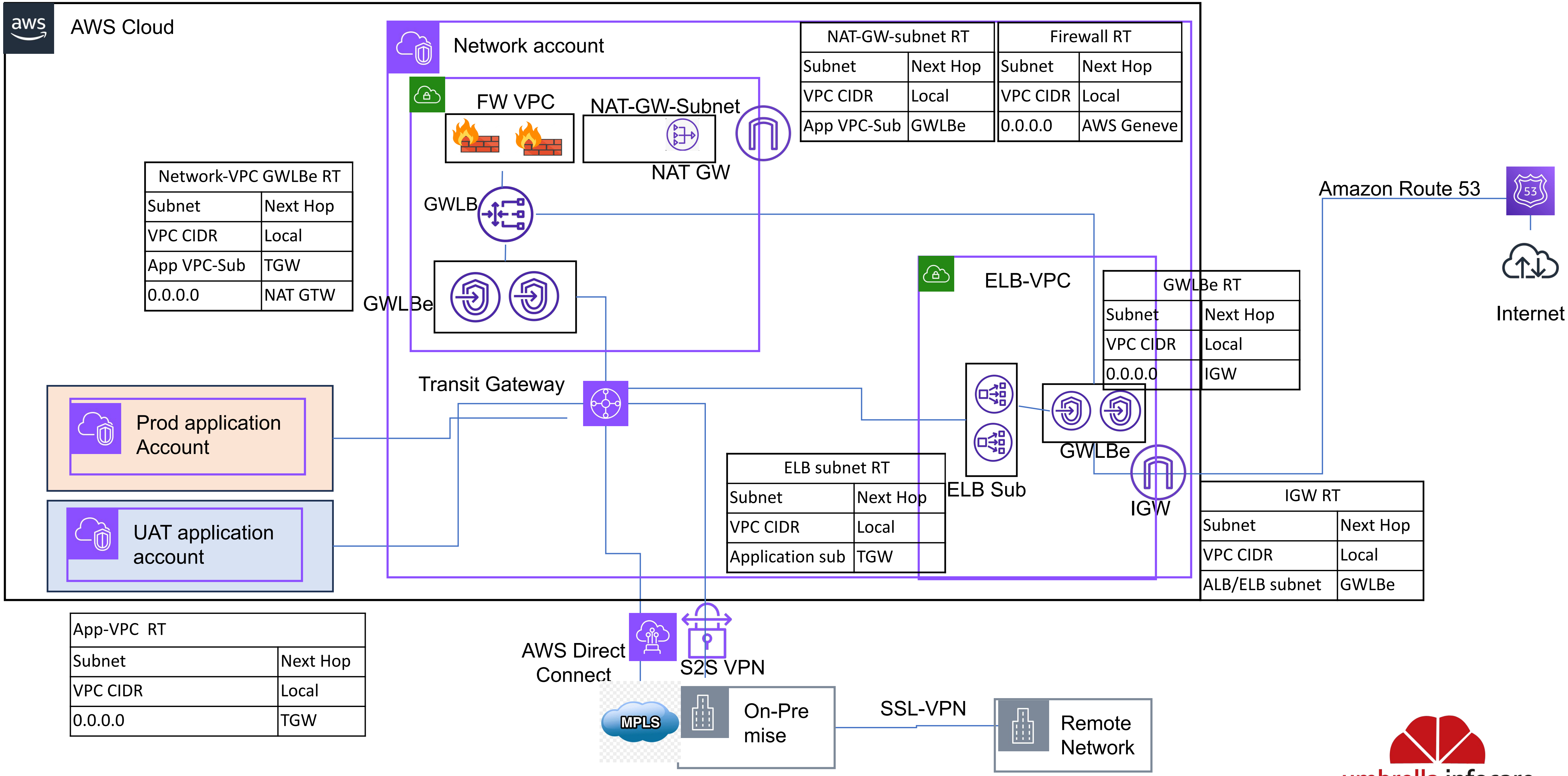
Route Table



NAT-GW-subnet RT	
Subnet	Next Hop
VPC CIDR	Local
App VPC-Sub	GWLBe

Firewall RT	
Subnet	Next Hop
VPC CIDR	Local
0.0.0.0	AWS Geneve

Network-VPC GWLBe RT	
Subnet	Next Hop
VPC CIDR	Local
App VPC-Sub	TGW
0.0.0.0	NAT GTW

GWLBe RT	
Subnet	Next Hop
VPC CIDR	Local
0.0.0.0	IGW

ELB subnet RT	
Subnet	Next Hop
VPC CIDR	Local
Application sub	TGW

IGW RT	
Subnet	Next Hop
VPC CIDR	Local
ALB/ELB subnet	GWLBe

App-VPC RT	
Subnet	Next Hop
VPC CIDR	Local
0.0.0.0	TGW

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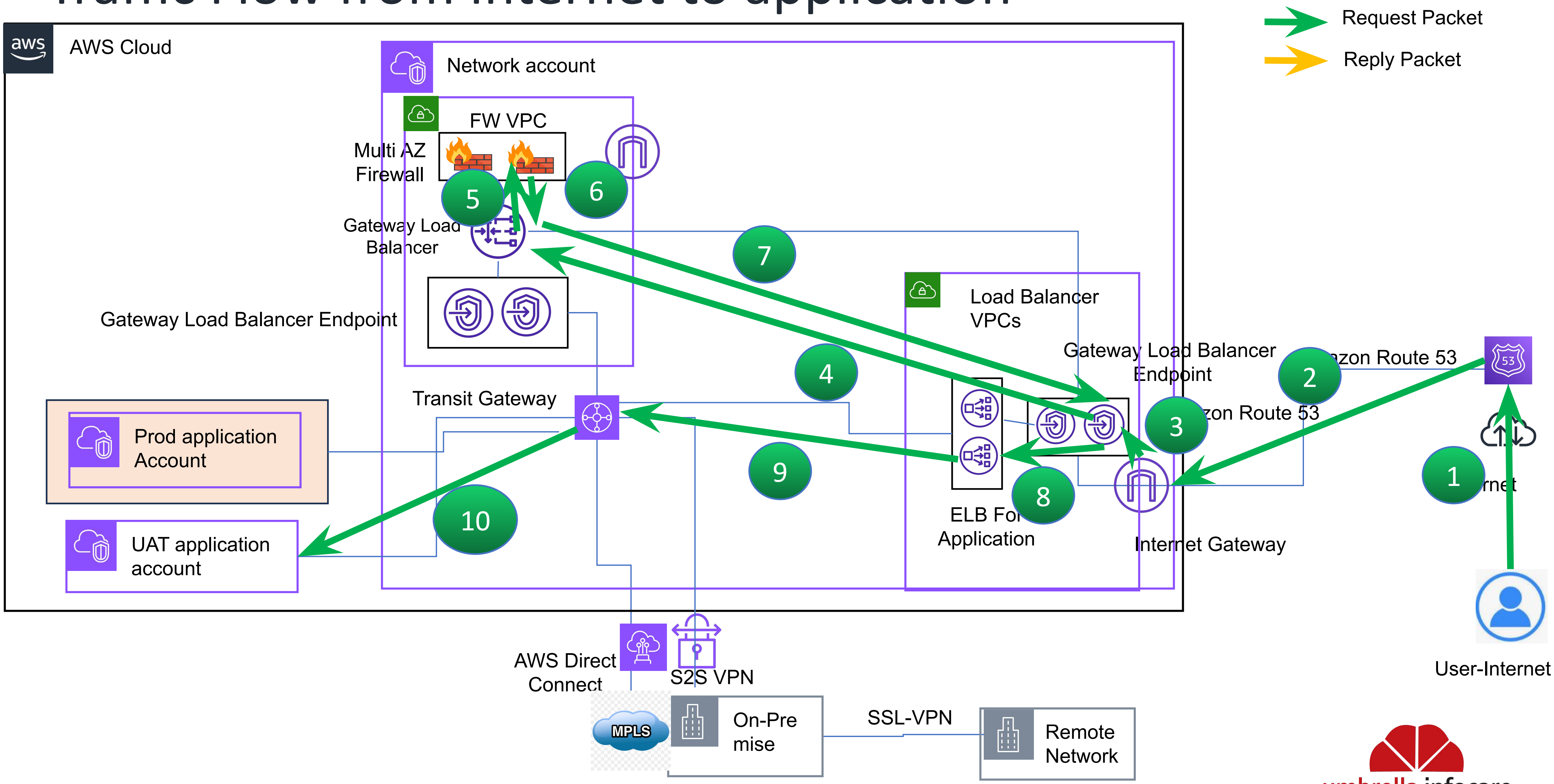
Traffic Flow from Internet to application



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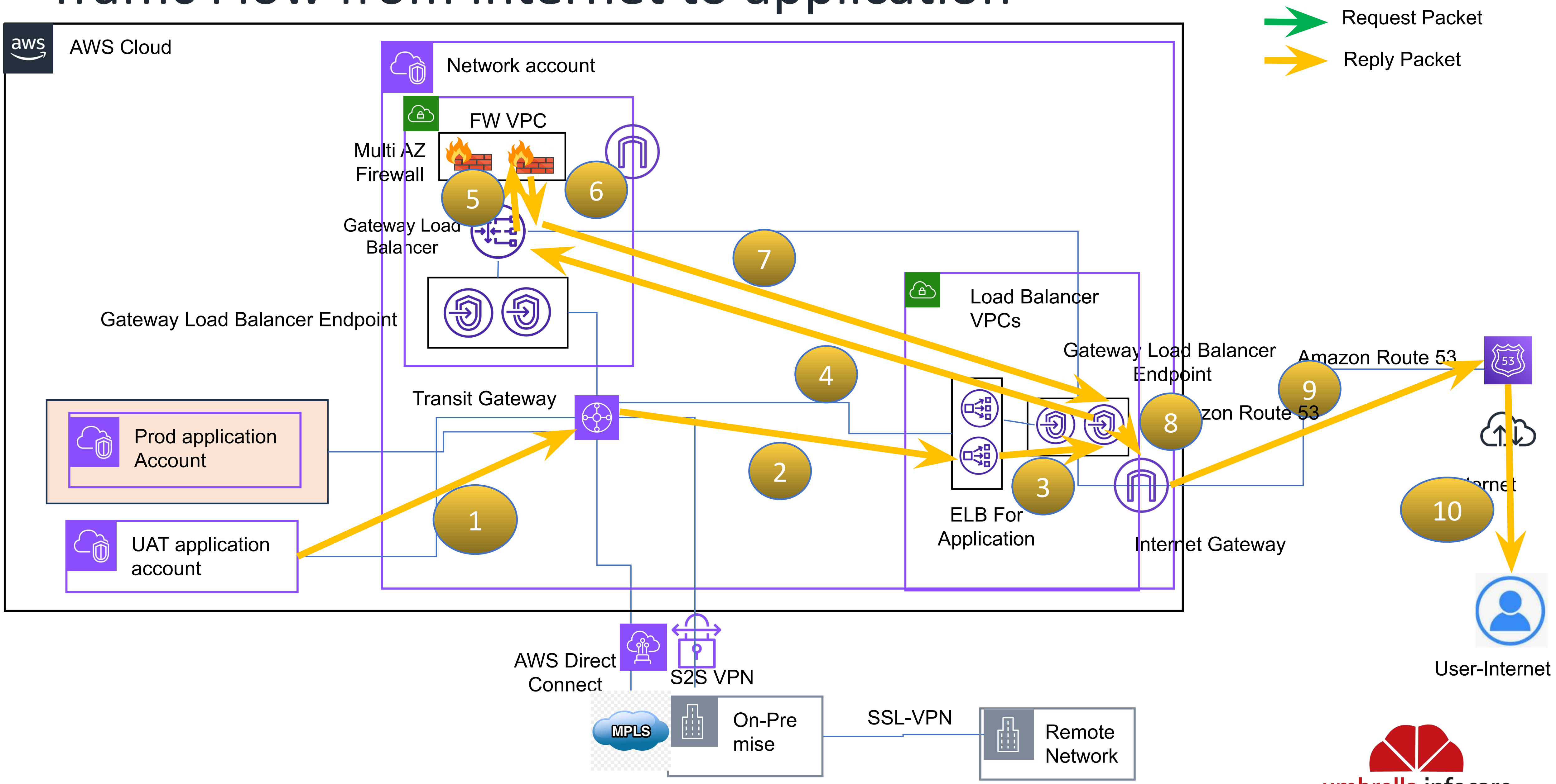
Traffic Flow from Internet to application



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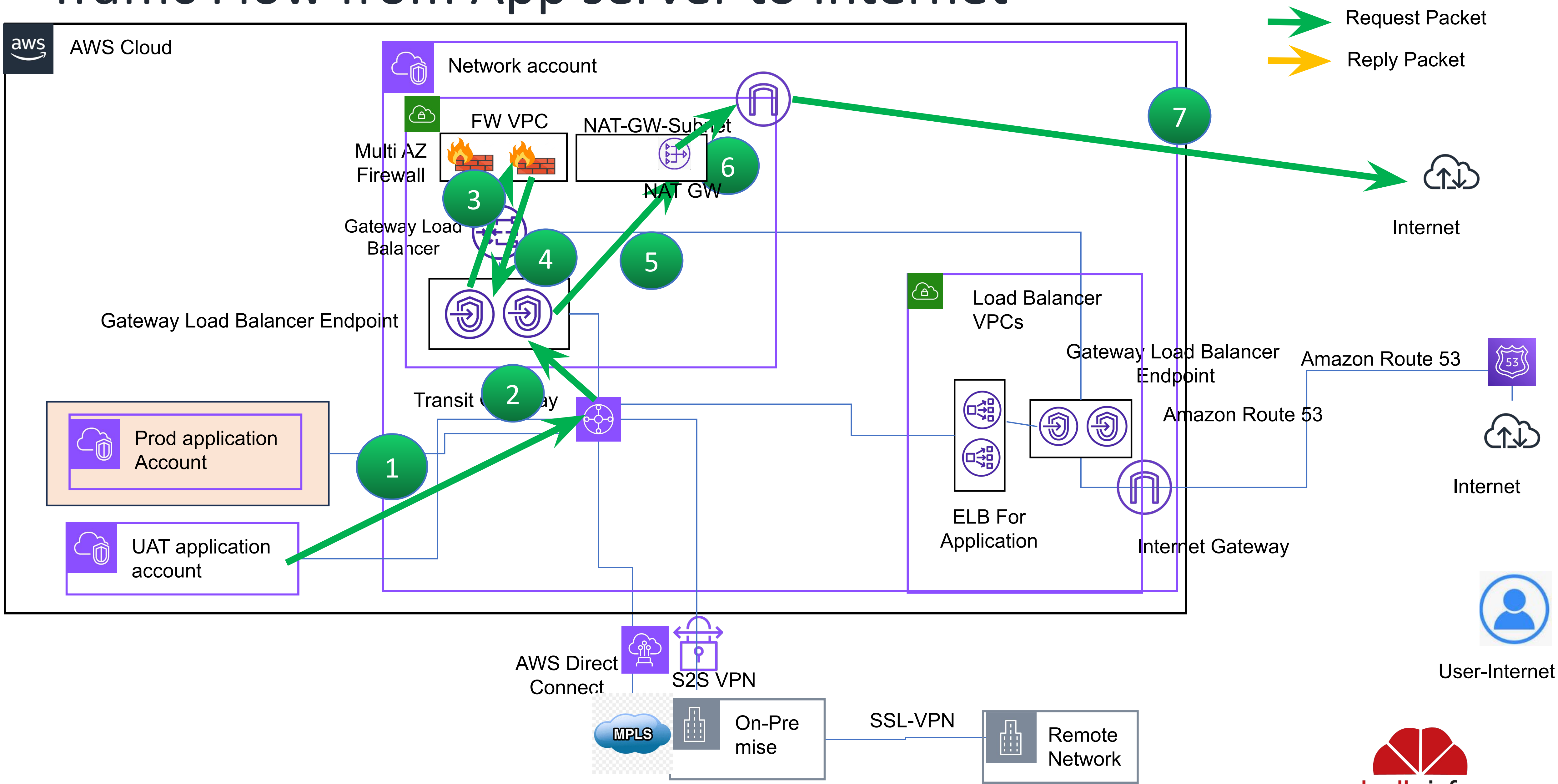
Traffic Flow from App server to Internet



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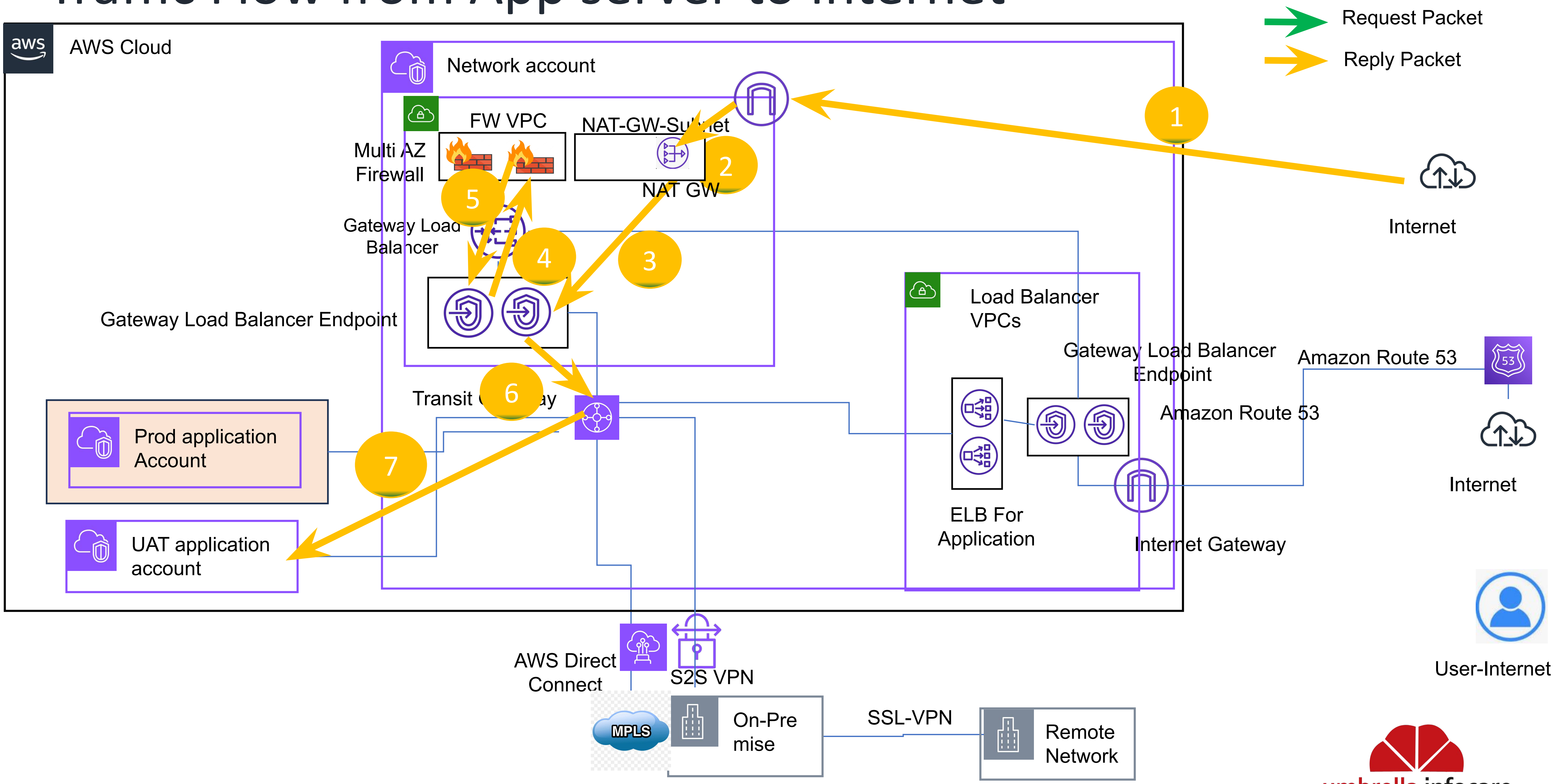
Traffic Flow from App server to Internet



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Architecture(ELB\GWLBe in dedicated VPC)

Deployment consideration

- A dedicated VPC will be used for deployment of Firewalls in Multi-AZ setup
- Firewall VPC will have gateway load balancer
- Firewall VPC will have gateway load balancer endpoints for each AZ
- A dedicated VPC will be used for deployment of Elastic Load Balancers and gateway load balancers.
- Gateway load balancer endpoint in firewall VPC will be used for outbound, VPN, direct connect and east-west traffic inspection.
- Gateway load balancer endpoint in Load balancer VPC will be used for inbound traffic inspection.
- Transit gateway in network account will be used for termination of direct connect & site to site VPN.
- Transit gateway will be used to communication between different VPCs and AWS accounts.
- Transit gateway will be used as the default gateway for all the instances in all application VPCs.
- Direct termination of Site to Site VPN and Remote access VPN is not feasibile in this architecture.
- Site to Site VPN can be terminated on transit gateway and can be inspected by Firewall.
- For remote access VPN a dedicate Firewall will be required(Small size firewall will be sufficient)

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Architecture(ELB\GWLBe in dedicated VPC)

North-South Traffic(Incoming)

- Traffic coming from internet will have DNS destination for Elastic load balancer(ELB) created in Network Account.
- Traffic destined for ELB, Internet Gateway will send the traffic to Gateway load balancer Endpoint using Edge association.
- Gateway load balancer endpoint will send to Gateway Load Balancer.
- Gateway load balancer will maintain the session and send to Firewall for inspection.
- After Inspection Firewall will send the traffic back to gateway load balancer and then to Gateway load balancer endpoint.
- Gateway load balancer endpoint will send the traffic to original destination, which is Elastic load balancer.
- Return Traffic will follow the same path.

Architecture(ELB\GWLBe in dedicated VPC)

North-South Traffic(Outgoing)

- Traffic originating from server will be send to transit gateway(Network Account) as default gateway
- Traffic going towards internet will reach Gateway load balancer endpoint(In Firewall VPC) after transit gateway
- Gateway load balancer endpoint will send to Gateway Load Balancer.
- Gateway load balancer will maintain the session and send to Firewall for inspection.
- After Inspection Firewall will send the traffic back to internet gateway of firewall vpc.
- Internet Gateway will send traffic to internet
- Return Traffic will follow the same path.

Architecture1(ELB\GWLBe in dedicated VPC)

Components Associated

- Firewall Quantity: 3
- Gateway load Balancer:1
- Gateway Load Balancer Endpoint: 4(considering 2 AZ architecture)
- Transit Gateway Attachment: 1 for each application VPC, firewall VPC, load balancer VPC, remote VPN VPC
- Internet Gateway: 3(Firewall VPC, ELB VPC, Remote VPN)

AWS Cost Associated

- Gateway load Balancer:1(Quantity)
  - $0.0133 per Gateway Load Balancer-hour (or partial hour) per AZ
  - $0.004 per GLCU-hour (or partial hour)

A GLCU measures the dimensions on which the Gateway Load Balancer processes your traffic (averaged over an hour). The three dimensions measured are:

  - New connections or flows: Number of newly established connections/flows per second.
  - Active connections or flows: Peak concurrent connections/flows, sampled minutely.
  - Processed bytes: The number of bytes processed by the load balancer in GBs.

Customer is charged only on one of the three dimensions that has the highest usage for the hour. A GLCU contains:

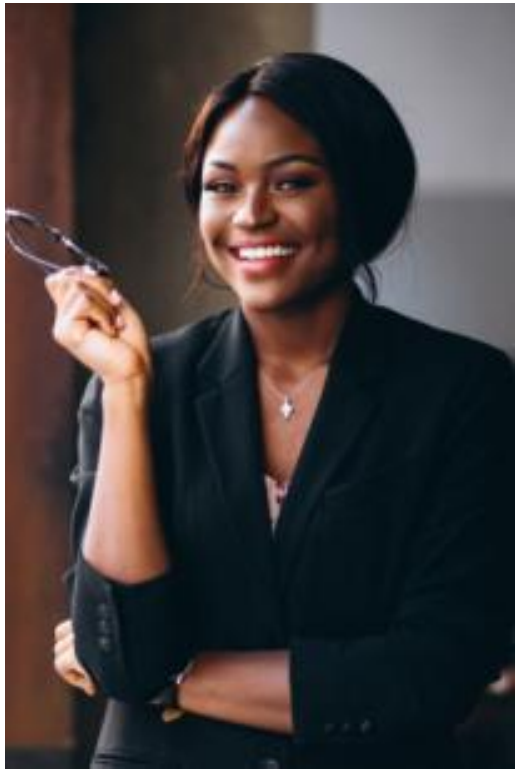
  - 600 new connections/flows per second.
  - 60,000 active connections/flows (sampled per minute).
  - 1 GB per hour for Amazon Elastic Compute Cloud (EC2) instances, containers, and IP addresses as targets.

- Gateway Load Balancer Endpoint: 2(1 for each AZ)
  - Pricing per VPC endpoint per AZ ($/hour) $0.013
  - Pricing per GB of Data Processed ($): $0.0035
- Transit Gateway
  - Price per AWS Transit Gateway attachment ($/hour): $0.07
  - Price per GB of data processed ($): $0.02

https://aws.amazon.com/privatelink/pricing/

https://aws.amazon.com/elasticloadbalancing/pricing/

https://aws.amazon.com/transit-gateway/pricing/



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